

Arduino Health Solutions

Heartbeat & ECG Monitor

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Importance of Monitoring

Heart Function

Heart rate and ECG provide insights into heart health.

Clinical Usage

Clinics use these metrics to identify potential heart issues.

Cost Challenges

Hospital monitors are accurate but often too expensive for many.

Wearable Demand

Wearable devices show the need for affordable monitoring solutions.

Project Objectives

Design

Create a cost-effective prototype that can measure both heartbeat and a single-lead ECG signal using off-the-shelf sensors and an Arduino microcontroller.

Measure

Capture heart rate, SpO_2, and ECG signals in real time and visualize them on a computer using the Arduino Serial Plotter or similar tools.

Integrate

Combine the MAX30102 optical heart rate and SpO_2 sensor with the AD8232 analog ECG front-end on a single Arduino UNO R4 WiFi platform.

Evaluate

Assess signal quality, stability, and responsiveness of the system, and identify where it behaves well and where it struggles compared to what we expect from medical devices.



System Overview

Heart Signals

Human subjects provide both optical signals (changes in blood volume at the fingertip) and electrical signals from the heart's conduction system.

AD8232 Module

The AD8232 module amplifies and filters a small ECG signal picked up by surface electrodes, producing a clean-enough analog waveform for the Arduino to sample.

MAX30102 Sensor

The MAX30102 uses red and infrared LEDs plus a photodiode to measure a photoplethysmography (PPG) signal, which we use to estimate heart rate and blood oxygen saturation.

Data Transmission

The Arduino UNO R4 WiFi reads both sensors, processes the data, and sends the values and waveforms to a computer for real-time plotting and analysis.

Hardware Components

Arduino

Arduino UNO R4 WiFi enables connectivity for our project.

MAX30102

The MAX30102 sensor measures heart rate and SpO2 levels.

AD8232

AD8232 is essential for capturing single-lead ECG signals.

Electrodes

Surface electrodes detect electrical activity from the body.

Prototyping

Bread board for prototyping wires to connect all components for system integration.

Circuit Connections

Wiring for Heartbeat and ECG Monitor

MAX30102

The MAX30102 sensor requires a VCC connection to 3.3V, a GND connection to ground, and SDA/SCL pins connected to the Arduino's I2C pins for communication.

AD8232

The AD8232 module can be powered with either 3.3V or 5V, needs a GND connection, and outputs data to an analog pin on the Arduino (e.g., A0).

Common Ground

It is essential to maintain a common ground across all components to ensure proper sensor operation and accurate signal readings throughout the monitoring system.

Software & Data Flow

Initialization

Initialize the Arduino and set up the sensor inputs.

Data Reading

Inside the main loop, the Arduino continuously reads PPG samples from the MAX30102 and analog voltage values from the AD8232, building a live stream of heart-related data..

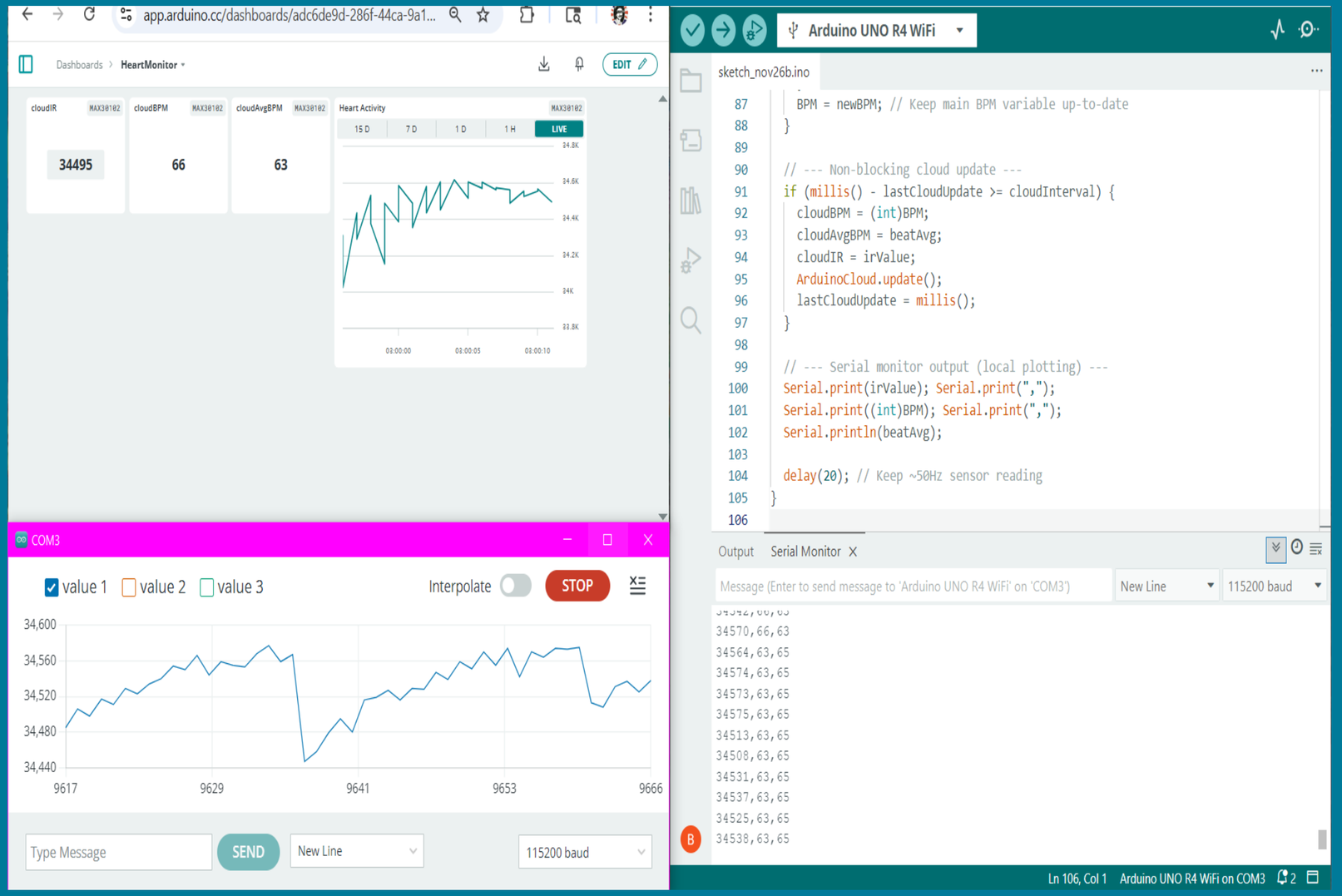
Noise Reduction

Applied a filter to reduce noise in the signals.

Peak Detection

The code identifies peaks in the PPG or ECG signal to estimate heart rate in beats per minute and sends both waveforms and calculated values to the serial monitor or plotter for visualization.

Results: Single-Lead ECG Signal Capture



Limitations & Challenges

Noise

Even small movements of the arms, torso, or electrode wires introduced large artifacts into the ECG signal, sometimes temporarily hiding the true waveform.

Pressure

For the MAX30102, pressing too hard or too lightly on the sensor changed the PPG amplitude and sometimes caused unstable or inaccurate heart rate and SpO_2 readings.

Filtering

Our prototype used only basic filtering and simple peak detection, so the system is more sensitive to noise and not comparable to medical-grade ECG machines in terms of reliability.

Single Lead Limitation

A single-lead ECG can show basic rhythm information but cannot provide the full spatial view of the heart’s activity that a standard 12-lead clinical ECG offers.

Future Improvements

Signal Filtering

Implementing digital band-pass and notch filters would help remove baseline wander and power-line interference, making ECG and PPG waveforms significantly cleaner.

Wireless Connectivity

Using the Arduino UNO R4's WiFi or adding Bluetooth support would enable wireless streaming of heart data to a phone app or web dashboard for more convenient monitoring.

Display Integration

Adding a small OLED or LCD screen would allow users to see heart rate and basic signal status directly on the device without needing a laptop.

Data Logging

Including SD card logging would make it possible to record longer sessions, analyze trends over time, and compare results under different conditions or activities.

Conclusion & Takeaways

Prototype Success

We **built a successful prototype** for heart rate monitoring.

Integration Experience

Integration of sensors with Arduino **was a valuable experience**.

Real-Time Acquisition

We gained insight into **real-time signal acquisition challenges**.

Prototype Limitations

Observing the gap between **prototypes and medical devices** was enlightening.